HMITiles

HMITiles
ISA-101–Inspired Open-Source HMI Tile Library for Industrial Dashboards

Overview
HMITiles is an open-source HMI (Human Machine Interface) tile library written in **B4X**, following widely accepted industrial HMI design principles.
It provides reusable, professional-grade HMI tiles for: Industrial dashboards, SCADA (Supervisory Control and Data Acquisition) front-ends, Machine and process HMIs

The focus of this library is **clarity, consistency, and disciplined HMI design -** not visual effects or UI decoration.

Design Goals
- Based on widely accepted industrial HMI design principles
- Alarm-first color discipline
- Clear and consistent state handling
- Touch-friendly layouts
- No animations or visual noise
- Minimal configuration required

This is a **non-commercial project** shared for inspiration.

B4J Example – Overview HMITiles

HMITiles Used

HMITileLabel (3)
HMITileDigitalClock
HMITileClock
HMITileButton (3)

HMITileSensor (3)
HMITileSeekBar (2)
HMITileGauge

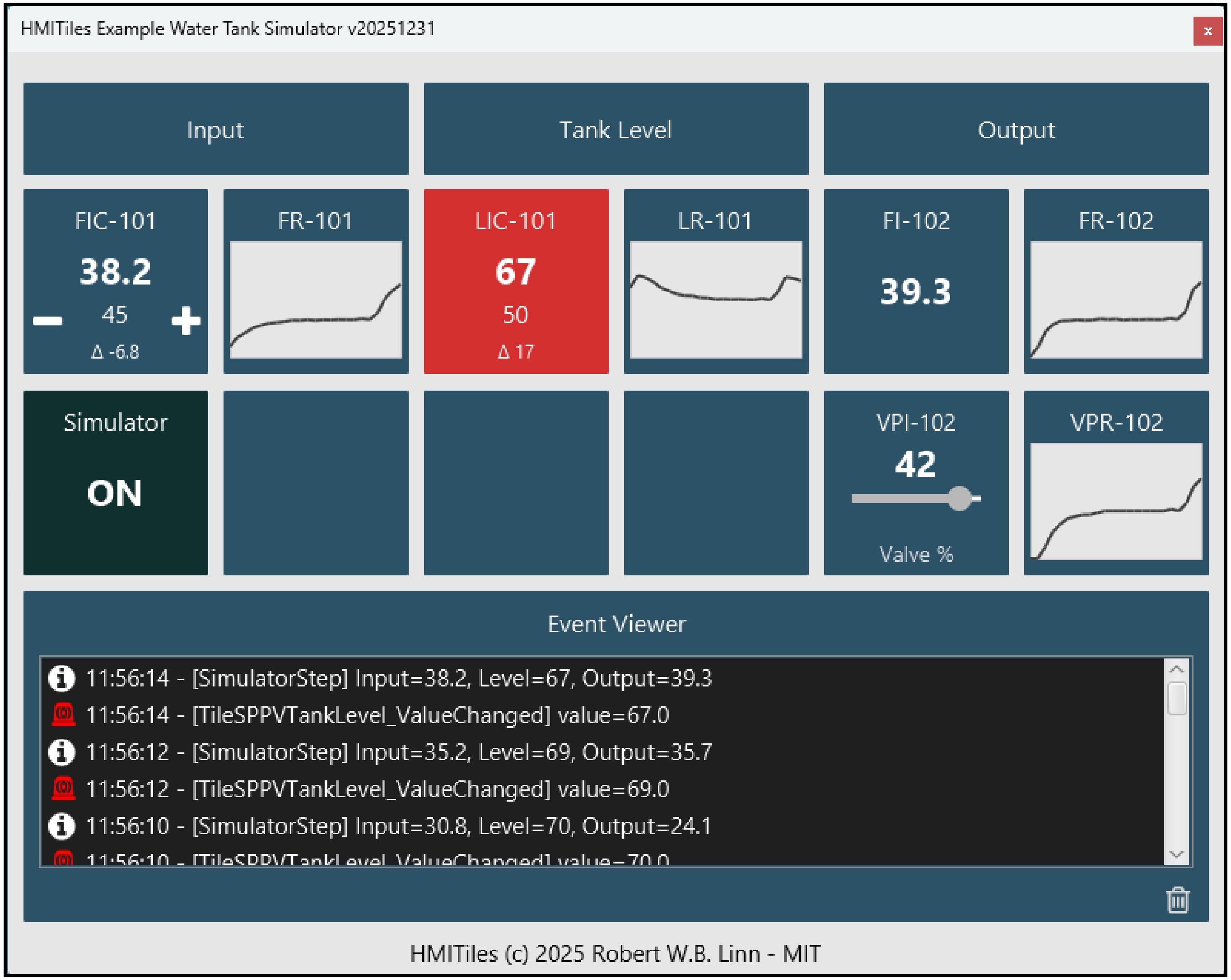
HMITileRGB (2)
HMITileLevel
HMITileReadOut
HMITileTrend
HMITileImage

HMITileSetPoint
HMITileSelectList
HMITileSelect
HMITileShape (8)
HMITileImageIcon

HMITileEventViewer
HMITileList

B4J Example - Water Tank Simulator

A simple interactive water tank PID simulator demonstrating ISA-101-style HMI design using HMITiles.



HMITiles Used

HMITileLabel

HMITileSetpoint
HMITileTrend
HMITileSetpoint
HMITileReadOut
HMITileTrend

HMITileButton
HMITileSeekBar
HMITileTrend

HMITileEventViewer

B4J Example – ArduinoLED

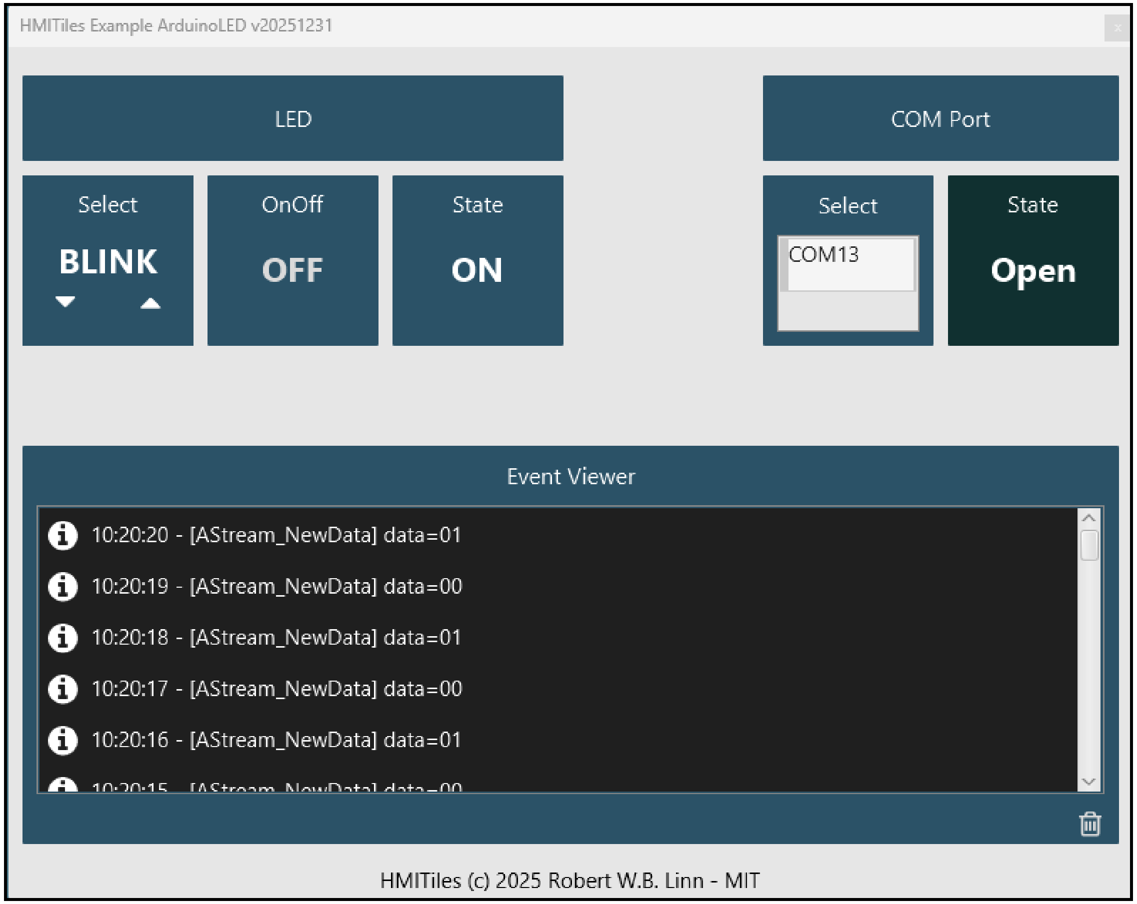
Control the state of an LED connected to an Arduino UNO via the serial line.

HMITiles Used

HMITileLabel

HMITileSelect
HMITileButton
HMITileReadOut

HMITileEventViewer



HMITiles Used

HMITileLabel

HMITileSelectList
HMITileButton

HMITileEventViewer

B4A Example – Overview

HMITiles Used

HMITileLabel (3)
HMITileDigitalClock
HMITileClock

HMITileButton (3)
HMITileTrend
HMITileGauge

HMITileSensor
HMITileReadOut
HMITileLevel

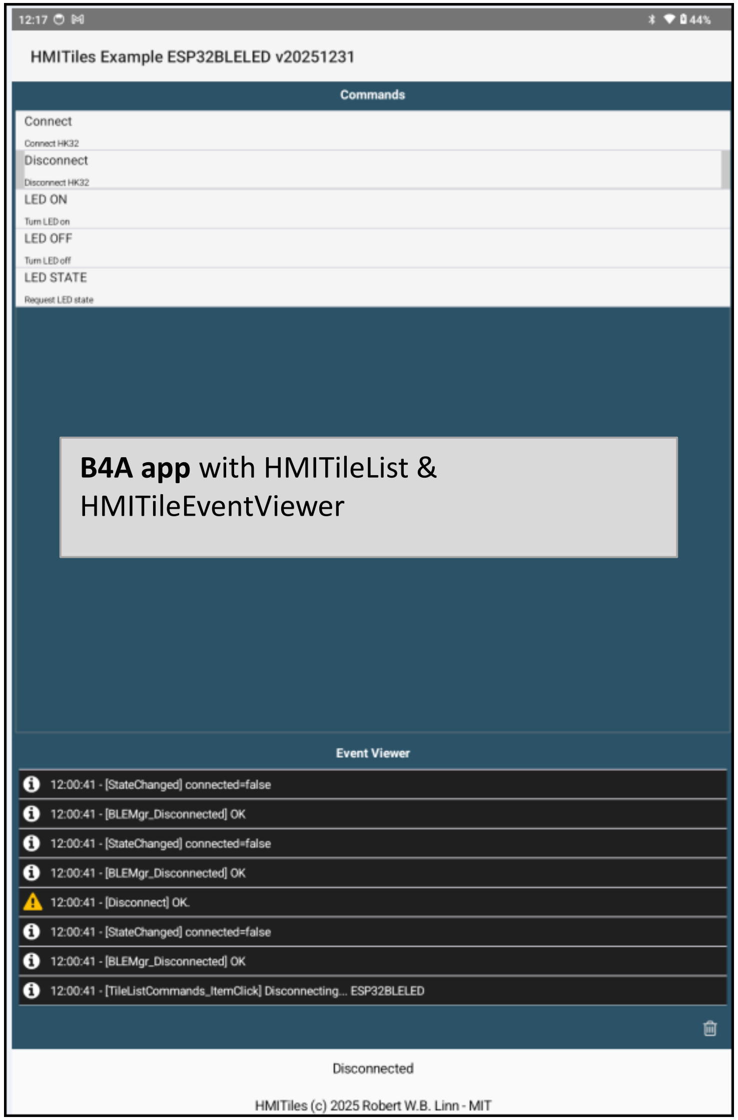
HMITileSeekBar (2)

HMITileSetPoint
HMITileSelectList
HMITileSelect

HMITileEventViewer

B4A Example – ESP32BLELED

Control the state of an LED connected to an ESP32 via the BLE (UART).

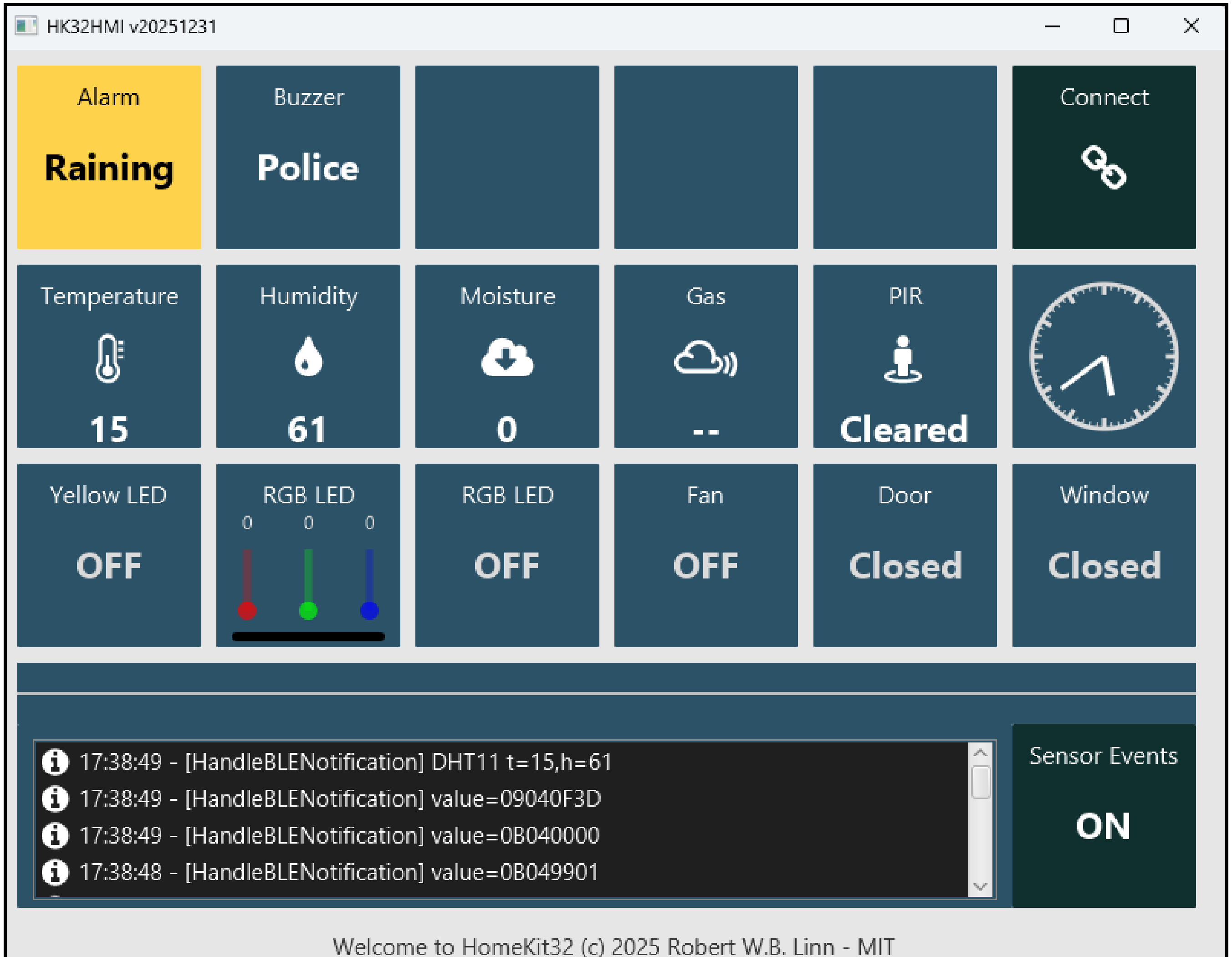


B4A app with HMITileList & HMITileEventViewer

```
[B4RBLEServer::onConnect] Client connected
[CommBLE.BLEServer_NewData] buffer HEX=010101
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010101
[CommBLE.BLEServer_Write] data=010101
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010101
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=010100
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010100
[CommBLE.BLEServer_Write] data=010100
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010100
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=0102
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=0102
[CommBLE.BLEServer_Write] data=010200
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010200
[B4RBLEServer::onWrite] Received bytes: 2
[B4RBLEServer::SetStartAdvertising] Started advertising: ESP32BLELED
[B4RBLEServer::onDisconnect] Client disconnected, restarting advertising
```

B4R program IDE log showing the commands: connect > command LED ON > command LED OFF > disconnect

HomeKit32 is a modular smart-home automation framework for the Keyestudio Smart Home Kit (KS5009), powered by **ESP32, BLE, MQTT and B4X**.



HMITiles Used

HMITileLabel (2)
HMITileButton

HMITileSensor (4)
HMITileDigitalClock
HMITileClock

HMITileButton (5)
HMITileRGB

HMITileEventViewer
HMITileButton

Screenshots